

Curriculum Vitae

Dr. Sitanshu Sekhar Sahu

Associate Professor
Department of Electronics and Communication Engineering
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Education

Ph.D.:

National Institute of Technology, Rourkela, India (2007-2011)

Dissertation: Analysis of Genomic and Proteomic Signals Using Signal Processing and Soft-computing Techniques

Bachelor of Engineering (B.E.):

Biju Patnaik University of Technology, Odisha, India (2001-2005)

Major: Electronics and Telecommunication Engineering

Research Experience

Postdoctoral Research Fellow,

Oklahoma State University, USA (Oct.2012-Oct.2014)

Graduate Research Scholar,

National Institute of Technology, Rourkela,India (2007-2011)

Visiting Graduate Research Scholar,

University of Western Ontario, Canada (Oct., 2008-March, 2009).

Academic Experience

Associate Professor, Department of Electronics and Communication Engineering, Birla Institute of Technology Mesra, Ranchi, India, August-2021-till date.

Assistant Professor, Department of Electronics and Communication Engineering, Birla Institute of Technology Mesra, Ranchi, India, September-2011-July 2021.

Guest Faculty, School of Electrical Sciences, Indian Institute of Technology Bhubaneswar, India, December, 2010- September, 2011.

Professional Activities

- Reviewer- Various Journals of Nature, IEEE, Elsevier Science, Springer, Taylor & Francis and BMC
- **Chaired a Technical session at** MidSouth Computational Biology and Bioinformatics Society Conference (MCBIOS), Oklahoma, USA held in April, 2014.
- **Chaired a Technical session at** International Conference on Nano-electronics, Circuits and Communication systems, Ranchi, Jharkhand, Indian held in May, 2015.
- **Chaired a Technical session at** International Conference on Data Science and Applications, MNIT Jaipur, Indian held in July, 2024.
- **Key note Speaker** at International Conference on Data Science and Applications, MNIT Jaipur, Indian held in July, 2024.
- **Member of Institute of Electrical and Electronics Engineers (IEEE), USA (Membership No: 90793764)**
- **Member of International Association of Engineers (IAENG), Hong Kong (Membership No: 289706)**

Fellowship and Achievements

- Received Postdoctoral Fellowship (2012-2014) from Oklahoma State University, USA.
- Received DFAIT-GSEP (October, 2008 – March, 2009) scholarship, Canadian Government.
- Received travel support from USDA, USA to attend a workshop on “Engineered Crops” from 28.04.2014 to 29.04.2014 at Des Moines, IA, USA.
- Got National Merit Scholarship from Govt. of India in 1995
- Got State Merit Scholarship from Govt. of Odisha in 1993
- Received Best paper award in the IEEE International Symposium of Asian Control Association on Intelligent Robotics and Industrial Automation (IRIA) Held at IIT Goa, India, 2021
- Jury Member, National wide Edge AI Innovation Challenge in Collaboration with DigiToad Technologies Pvt. Ltd. and ST Microelectronics, 2024

Primary Research Interest

- Bio-Signal Processing
- Image Processing
- Artificial Intelligence and Machine Learning
- Embedded AI
- Bioinformatics
- Computer Vision

List of Publications

International Journals

1.	Pandit Vivek Kumar Pandey, Sitanshu Sekhar Sahu, "Multilingual Voice Signal Analysis for Parkinson's Disease Detection Using Instance-Based Domain Adaptation through Kernel Mean Matching" <i>Computer Speech & Language</i> , 102010, May 2026
2.	Pandit Vivek Kumar Pandey, Sitanshu Sekhar Sahu, "Parkinson's Disease Detection Using Hybrid Siamese Neural Network and Support Vector Machine in Multilingual Voice Signal" <i>Journal of Voice</i> , 4:S0892-1997(25)00261-9, Aug. 2025
3.	Pandit Vivek Kumar Pandey, Sitanshu Sekhar Sahu, "Cascade Convolutional Neural Network-Autoencoder Method for Parkinson's Disease Detection Using Multilingual Voice Signal" <i>SN Computer Science</i> , Vol 6 (802), pp:1-12, Sept 2025
4.	Thakur, A., Mishra, S.K., Bhutani, M., P.P. Dash, P.K. Upadhyay, S. S. Sahu, "Deep learning approaches for detecting malignant melanoma in dermoscopic imagery" <i>International journal of Information Technology</i> (July 2025).
5.	Tuhin Mukherjee, Ashok Pattnaik, Sitanshu Sekhar Sahu, "Resonant recognition model-guided biophysical insights into POMC, leptin, adiponectin, BDNF, insulin, irisin, GDF-15 interactions & executing PBMT targeting leptin/BDNF circuit to attenuate obesity" <i>Lasers in Medical Science</i> , Vol. 40:288, June 2025
6.	Pandit Vivek Kumar Pandey, Sitanshu Sekhar Sahu, Biswajit Karan, Sudhansu Kumar Mishra "Parkinson's Disease Prediction using CNN-LSTM model from Voice Signal", <i>SN Computer Science</i> , Vol 5 (4), pp:381, March 2024
7.	Rupesh Kumar Sinha, Sitanshu Sekhar Sahu, "Multi-level Image Security using Modified Rubik's Cube Algorithm" <i>Int. J. of Information and Computer Security</i> , Vol 23 (4), pp: 396-409, June 2024
8.	Tuhin Mukherjee, Ashok Pattnaik, Sitanshu Sekhar Sahu, "Analyzing VEGFA/VEGFR1 Interaction: Application of the Resonant Recognition Model-Stockwell Transform Method to Explore Potential Therapeutics for Angiogenesis-Related Diseases", <i>The Protein Journal</i> , 43, 697–710, July 2024
9.	Thakur, A., Kumar, A., Mishra, S.K., Subhendu Kumar Behera, Jagannath Sethi, Sitanshu Sekhar Sahu, Subrat Kumar Swain, Product Length Predictions with Machine Learning: An Integrated Approach Using Extreme Gradient Boosting. <i>SN COMPUT. SCI.</i> 5 , 659 June (2024)
10.	Biswajit Karan, Satyajit Mahapatra, Sitanshu Sekhar Sahu, Dev Mani Pandey, Sumit Chakravarty, "Computational Models for Prediction of Protein-Protein Interaction in Rice and Magnaporthe grisea" <i>Frontiers in Plant Science</i> , Vol. 13, PP: 5656, Feb. 2023 (SCIE), ISSN: 1664-462X
11.	Amit Kumar Sahoo, Sudhansu Kumar Mishra, Deep Shekhar Acharya, Sitanshu Sekhar Sahu, Sanchita Paul, Vikash Kumar Gupta, "Identification and control of Maglev system using fractional and integer order PID controller" <i>Journal of Intelligent & Fuzzy Systems</i> , vol. 44, no. 5, pp. 7277-7289, May 2023 (SCIE)
12.	Biswajit Karan, Sitanshu Sekhar Sahu, Juan Rafael Orozco-Arroyave, "An investigation about the relationship between dysarthria level of speech and the neurological state of Parkinson's patients", <i>Biocybernetics and Biomedical Engineering</i> , Vol 42(2), pp:710-726, June 2022 (SCIE), ISSN: 0208-5216
13.	Vikas Pathak, Satyasai Jagannath Nanda, Amit Mahesh Joshi, Sitanshu Sekhar Sahu, "Identification of characteristics frequency and hot-spots in protein sequence of COVID-19 disease", <i>Biomedical Signal Processing and Control</i> , Vol. 78, PP: 103909, Sept. 2022 (SCIE)

	ISSN: 1746-8094
14.	Satyajit Mahapatra, Sitanshu Sekhar Sahu, "ANOVA-particle swarm optimization-based feature selection and gradient boosting machine classifier for improved protein–protein interaction prediction", <i>Proteins: Structure, Function, and Bioinformatics</i> , 90(2):443-454, Sept. 2021 (SCIE), ISSN: 1097-0134
15.	Vikas Pathak, Satyasai Jagannath Nanda, Amit Mahesh Joshi, Sitanshu Sekhar Sahu, "FPGA implementation of high-speed tunable IIR band pass notch filter for identification of hot-spots in protein", <i>International Journal of Circuit Theory and Applications</i> , Vol. 19(11), pp: 3748-3765 Sept. 2021 (SCIE), ISSN: 0098-9886
16.	Satyajit Mahapatra, Sitanshu Sekhar Sahu, "Improved prediction of protein-protein interaction using hybrid of functional-link Siamese neural network and gradient boosting machines", <i>Briefings in Bioinformatics</i> , Vol. 22(6), Nov. 2021(SCIE), ISSN: 1477-4054
17.	Biswajit Karan, Sitanshu Sekhar Sahu," An improved framework for Parkinson's Disease Prediction using Variational Mode Decomposition-Hilbert Spectrum of speech signal", <i>Biocybernetics and Biomedical Engineering</i> , vol. 41 (2), pp: 717-732, April 2021(SCIE), ISSN: 0208-5216
18.	Satyajit Mahapatra, Sitanshu Sekhar Sahu, "Integrating Resonant Recognition Model and Stockwell Transform for Localization of Hotspots in Tubulin" <i>IEEE Transactions on NanoBioscience</i> , Vol. 20 (3), July, 2021 (SCIE), ISSN: 1536-1241
19.	Biswajit Karan, Sitanshu Sekhar Sahu, Juan Rafael Orozco-Arroyave and Kartik Mahto, " Non-negative matrix factorization-based time-frequency feature extraction of voice signal for Parkinson's disease prediction," <i>Computer Speech and Language</i> , Vol.69, 101216, Sept. 2021 (SCIE Indexed), ISSN: 0885-2308
20.	Satyajit Mahapatra, Vivek Raj Gupta, Sitanshu Sekhar Sahu, Ganapati Panda, "Deep neural network and extreme gradient boosting based Hybrid classifier for improved prediction of Protein-Protein interaction" <i>IEEE/ACM Transactions on Computational Biology and Bioinformatics</i> , Vol. 19 (1), Feb. 2021 (SCIE Indexed), ISSN: 1545-5963
21.	Ankita Mishra, Sitanshu Sekhar Sahu, Rajeev Sharma, Sudhansu Kumar Mishra, "Denoising of Electrocardiogram Signal Using S-Transform Based Time–Frequency Filtering Approach" <i>Arabian Journal for Science and Engineering</i> , Vol. 46, pp: 9515–9525, Feb. 2021 (SCIE Indexed) ISSN: 2191-4281
22.	Priya Sagar, Ashruti Upadhyaya, Sudhansu Kumar Mishra, Rudra Narayan Pandey, Sitanshu Sekhar Sahu, Ganapati Panda, "A Circular Adaptive Median Filter for Salt and Pepper Noise Suppression from MRI Images " <i>Journal of Scientific & Industrial Research</i> , Vol. 79, pp. 941-944, October, 2020 (SCIE Indexed), ISSN: 0975-1084
23.	Vikas Pathak, Satyasai Jagannath Nanda, Amit Mahesh Joshi, Sitanshu Sekhar Sahu, "Hardware implementation of infinite impulse response anti-notch filter for exon region identification in eukaryotic genes" <i>International Journal of Circuit Theory and Applications</i> , vol. 48(12), pp: 2242-2256, July 2020 (SCIE Indexed), ISSN: 0098-9886
24.	Biswajit Karan, Sitanshu Sekhar Sahu, Juan Rafael Orozco-Arroyave, Kartik Mahto, "Hilbert spectrum analysis for automatic detection and evaluation of Parkinson's speech" <i>Biomedical Signal Processing and Control</i> , vol. 61, August 2020 (SCIE Indexed), ISSN: 1746-8094
25.	Shikha Prakash, Sitanshu Sekhar Sahu, "Soil Moisture Prediction Using Shallow Neural Network" <i>International Journal of Advanced Research in Engineering and Technology</i> , Vol. 11, 6, pp: 426-435, June 2020 (SCOPUS Indexed), ISSN 2350-0328
26.	Vikas Pathak, Satyasai Jagannath Nanda, Amit Mahesh Joshi, Sitanshu Sekhar Sahu, "VLSI implementation of anti-notch lattice structure for identification of exon regions in Eukaryotic genes" <i>IET Computers & Digital Techniques</i> , vol. 14 (5), pp: 217-229 Sept. 2020. (SCIE

	Indexed) ISSN: 1350-2387
27.	Biswajit Karan, Sitanshu Sekhar Sahu , Kartik Mahto, "Parkinson disease prediction using intrinsic mode function based features from speech signal" <i>Biocybernetics and Biomedical Engineering</i> , vol. 40, pp: 249-264, April 2020, (SCI Indexed) ISSN: 0208-5216
28.	Sitanshu S. Sahu, Cristian D. Loaiza , Rakesh Kaundal, "Plant-mSubP: a computational framework for the prediction of single and multi-target protein subcellular localization using integrated machine-learning approaches" <i>AoB PLANTS</i> , vol. 12, no. 3, pp: plz068, Dec. 2019 (SCIE Indexed) ISSN: 2041-2851
29.	Navneet Nayan, Sitanshu Sekhar Sahu, Sanjeet Kumar, "Detecting anomalous crowd behavior using correlation analysis of optical flow", <i>Signal, Image and Video Processing</i> , 13(6), pp: 1233-1241, 2019, https://doi.org/10.1007/s11760-019-01474-9 (SCI Indexed), ISSN: 18631711
30.	Rupesh Kumar Sinha and S.S. Sahu, "Adaptive firefly algorithm based optimized key generation for image security" <i>Journal of Intelligent & Fuzzy Systems</i> , Vol. 36, no. 5, pp: 4437-4447, 2019, DOI:10.3233/JIFS-169998, (SCI Indexed) ISSN: 1064-1246
31.	S Kumari, SS Sahu, B Gupta, " Efficient SSHI circuit for piezoelectric energy harvester uses one shot pulse boost converter" <i>Journal of Analog Integrated Circuits and Signal Processing</i> , vol.97, pp.545-555, 2018 (SCI Indexed) (ISSN: 0925-1030) https://doi.org/10.1007/s10470-018-1218-1
32.	Rupesh Kumar Sinha and Sahu S.S, "Secure Image Encryption Based on Improved Bat Optimized Piecewise Linear Chaotic Map through Integrated Permutation-Confusion and Double diffusion", <i>Journal of Intelligent & Fuzzy Systems</i> , Vol. 35, no. 2, pp: 1567-1578, 2018 (SCI Indexed) (ISSN: 1064- 1246), ISSN: 1064-1246
33.	Ram Chandra Barik, Sitanshu Sekhar Sahu and SuvamoyChangder, "A Novel Smooth texture based visual cryptography approach for secure communication" <i>Journal of Electronic Security and Digital Forensic</i> , Vol. 10, No. 2, pp: 109-137, 2018 (SCOPUS Indexed) (ISSN:1751-9128)
34.	Jai Utkarsh, Raju Kumar Pandey, Shrey Kumar Dubey, Shubham Sinha, S. S. Sahu, "Classification of Atrial Arrhythmias using Neural Networks", <i>IAES International Journal of Artificial Intelligence</i> , Vol. 7, No. 2, pp. 90-94, 2018, (SCOPUS Indexed), DOI: http://doi.org/10.11591/ijai.v7.i2 . (ISSN:2089-4872)
35.	Chandan Pandey, Rokkam Sandeep, AikanshPriyam, SatyajitMahapatra, Sitanshu S. Sahu , "Predicting Protein-RNA Interaction Using Sequence Derived features and Machine Learning Approach" <i>International Journal of Data Mining and Bioinformatics</i> , Vol. 19, issue.3, pp: 270-282, 2017 (SCIE Indexed) (ISSN:1748-5681)
36.	M. Kumar, S.K. Mishra and S.S. Sahu , "Cat Swarm Optimization based Functional Link Artificial Neural Network Filter for Gaussian Noise Removal from Computed Tomography Images" <i>Applied Computational Intelligence and Soft Computing</i> , Volume 2016, pp:1-6, Article ID 6304915. (SCOPUS Indexed) ISSN: 1687-9724
37.	Nithin V George, Kristy F Tiampo, Sitanshu S. Sahu , Stephane Mazzotti, LaluMansinha, Ganapati Panda" Identification of Glacial Isostatic Adjustment in Eastern Canada Using S Transform Filtering of GPS Observations" <i>Journal of Pure and Applied Geophysics</i> , vol. 169(8), pp: 1507-1517, 2012. (SCI Indexed) ISSN: 0033-4553
38.	Sitanshu S. Sahu and Ganapati panda" Efficient Localization of Hot Spot in Proteins Using A Novel S-Transform Based Filtering Approach" <i>IEEE /ACM Transaction on Computational Biology and Bioinformatics</i> , vol. 8, issue 5, pp. 1235-1246, 2011 (SCIE Indexed) ISSN: 1545-5963
39.	Sitanshu S. Sahu and Ganapati panda" Identification of Protein Coding Regions In DNA Sequence Using a Time-Frequency Filtering Approach" <i>Journal of Genomics, Proteomics and Bioinformatics</i> , vol. 9(1-2), pp. 45-55, 2011. (SCI Indexed) ISSN: 1672-0229
40.	Sitanshu S. Sahu and Ganapati panda" A Novel Feature Representation Method based Chou's Pseudo Amino Acid Composition for Protein Structural Class Prediction" <i>Journal of</i>

	<i>Computational Biology and Chemistry</i> , vol. 34, issues 5-6, pp. 320-327, 2010 (SCI Indexed) ISSN: 1476-9271.
41.	Sitanshu S. Sahu and G. Panda, “A DSP Approach for Protein Coding Region Identification In DNA Sequences”, <i>International Journal of Signal and Image Processing</i> , Vol. 1, Issue 2, pp. 75-79, March 2010.

International Conference proceedings

1.	Anshu Mala Kispotta, Rupesh Kumar Sinha, Sitanshu Sekhar Sahu, “AI-Powered Diagnosis of Voice Disorders Using Mel-Frequency Feature and Machine Learning” IEEE International Conference on Computer Vision and Machine Intelligence, Oct. 2025
2.	Atul Arya, Sachin Kumar Singh, Richa Mishra, Vimal Kumar Singh Yadav, Sitanshu Shekhar Sahu, “Direct Writing of Silver Nanoparticle Patterns Through Liquid Microprinting” IEEE 7th International Conference on Emerging Electronics (ICEE), Dec. 2025
3.	Pandit Vivek Kumar Pandey, Sitanshu Sekhar Sahu, Biswajit Karan, Juan Rafael Orozco-Arroyave, “Parkinson’s Disease Detection Using Wavelet Packet Absolute Amplitude Deviation (WPAAD) from voice signals” IEEE International Conference on Acoustics, Speech, and Signal Processing, June 2025
4.	Pandit Vivek Kumar Pandey, Sitanshu Sekhar Sahu, Sumit Chakravarty, Craig Chin, “CNN-LSTM-Based Lung Sound Analysis for Pneumonia Detection” IEEE SoutheastCon 2025, pp: 1072-1077, March 2025
5.	Pandit Vivek Kumar Pandey, Sitanshu Sekhar Sahu, “Hybrid CNN-Bottleneck Encoder Based Feature Extraction Technique for Parkinson’s Disease Detection from Voice Signal” International Conference on Data Science and Applications, vol. 1237, pp: 351-361, May 2025
6.	Pandey, P.V., Karan, B., and Sahu, S.S., “Parkinson’s disease Prediction using Intrinsic Mode Function Instantaneous Amplitude Deviation” National Conference on Intelligent Systems, IoT, and Wireless Communication for the Society, Feb 2024
7.	Pandit Vivek Kumar Pandey, Sitanshu Sekhar Sahu, “Speech Signal Analysis Using Hybrid Feature Extraction Technique for Parkinson’s Disease Prediction”, International Conference on Data science and Applications, 15 July 2023
8.	Mrinalini Barik, Sitanshu Sekhar Sahu, Rupesh Kumar Sinha, “IoT based soil health monitoring system using soil moisture sensor and PH sensor” International conference on Data Science and Intelligent Applications, April 28-29, 2023
9.	Garima Thakur, Sitanshu Sekhar Sahu, Ying Xie, Sumit Chakravarty, “Classification of Heart Sound using Tunable Q Wavelet Transform and Machine Learning” IEEE International Conference on Big Data, Japan, pp: 2800-2803, Dec 2022
10.	Rohith Sai V, Biswajit Karan, Garima Thakur, Ashutosh Rath, Sitanshu Sekhar Sahu, “Heart Sound Abnormality Detection using Wavelet Packet Features and Machine Learning”, IEEE International Symposium of Asian Control Association on Intelligent Robotics and Industrial Automation (IRIA), pp:310-314, Sept. 2021
11.	Vikas Pathak, Satyasai Jagannath Nanda, Amit Mahesh Joshi, Sitanshu Sekhar Sahu, “FPGA Implementation of High Speed Anti-notch Lattice filter for Exon Region Identification in Eukaryotic Genes” IEEE Computer Society Annual Symposium on VLSI, August 2021, pp: 418-42, ISBN:978-1-6654-3947-3
12.	Vikas Pathak, Satyasai Jagannath Nanda, Amit Mahesh Joshi, Sitanshu Sekhar Sahu, “VLSI Implementation of Tunable Band-Pass Notch IIR Filter for Localization of Hot spots in Proteins”, International Conference on Microelectronics, Computing and Communication Systems, Springer, Singapore, pp: 563-575, March 2021

13.	Vartika V., Singh S., Das S., Mishra S.K., Sahu S.S., "A Review on Intelligent PID Controllers in Autonomous Vehicle" Advances in Smart Grid Automation and Industry 4.0., Lecture Notes in Electrical Engineering, Springer, Singapore, Vol 693, April, 2021
14.	Biswajit Karan, Sitanshu Sekhar Sahu, Kartik Mahto, "Stacked auto-encoder based Time-frequency features of Speech signal for Parkinson disease prediction" IEEE International Conference on Artificial Intelligence and Signal Processing, pp: 1-4, 2020
15.	Satyajit Mahapatra, Sitanshu Sekhar Sahu, "Boosting predictions of Host-Pathogen protein interactions using Deep neural networks", IEEE International Students' Conference on Electrical, Electronics and Computer Science (SCEECS), pp:1-4, Feb. 2020
16.	Satyajit Mahapatra, Anish Kumar, Animesh Sharma, Sitanshu Sekhar Sahu, "Effect of Dimensionality Reduction on Classification Accuracy for Protein-Protein Interaction Prediction" Advanced Computing and Intelligent Engineering, Springer, pp: 3-12, 2020
17.	Vishnu Vansh, Kumar Chandrasekhar, CR Anil, Sitanshu Sekhar Sahu, "Improved Face Detection Using YCbCr and Adaboost" Computational Intelligence in Data Mining, Springer, pp: 689-699, 2020
18.	Rupesh Kumar Sinha, Iti Agrawal, Kritika Jain, Anushka Gupta, SS Sahu, "Image Encryption Using Modified Rubik's Cube Algorithm" Advances in Computational Intelligence, Springer, pp: 69-78, 2020
19.	Biswajit Karan, Satyajit Mahapatra, Sitanshu Sekhar Sahu, "Prediction of Protein Interactions in Rice and Blast Fungus Using Machine Learning" IEEE International Conference on Information Technology, pp: 33-36, 2019
20.	Sweta Kumari, Subrat Kumar Swain, SitanshuSekhar Sahu, Aditya Kumar, Prashant Kumar, Bharat Gupta, "A Closed loop robust controller for SSHI based piezoelectric energy harvester" Annual Conference on Computer-Aided Developments in Electronics and Communication, CRC Press, pp:113, March 2019.
21.	Anu Sri, Sitanshu Sekhar Sahu, "Improved fractal-SPIHT hybrid image compression algorithm" IEEE International Conference on Computing, Communication and Networking Technologies, pp: 1-4, July 2019
22.	Biswajit Karan, Kartik Mahto and Sitanshu Sekhar Sahu, "Detection of Parkinson Disease using Variational Mode Decomposition of the Speech Signal" IEEE International Conference on Communication and Signal Processing, April 3-5, 2018.
23.	Vikas Pathak, Satyasai Jagannath Nanda, Amit Mahesh Joshi, Sitanshu Sekhar Sahu, "High Speed Implementation of Notch/Anti-notch IIR Filter on FPGA", Dec 18-20, 2018
24.	Rupesh Kumar Sinha, Baddigam Asha, Niraj San, Savvy Prasad, S.S.Sahu, "Chaotic Image Encryption Scheme Based on Modified Arnold Cat Map and Henon Map", IEEE International Conference on Current trends towards Converging Technologies (ICCTCT 2018), Coimbatore, India, March 1-3,2018
25.	Khushboo Kumari, Sitanshu Shekhar Sahu, Rupesh Kumar Sinha, "R Peak Detection Using Empirical Mode Decomposition With Shannon Energy Envelope" International Conference on Inventive Communication and Computational Technologies, 2018
26.	Rupesh Kumar Sinha, Baddigam Asha, Niraj San, Savvy Prasad, S.S.Sahu, "Chaotic Image Encryption Scheme Based on S-Box Substitution " International Conference on Inventive Research in Computing Applications, 2018
27.	Shikha Prakash Animesh Sharma Sitanshu Shekhar Sahu, "Soil moisture prediction using machine learning" International Conference on Inventive Communication and Computational Technologies, 2018
28.	Apurva Sriwastwa, Shikha Prakash, Mrinalini, Swati Swarit, Khushboo Kumari, Sitanshu Sekhar Sahu, "Detection of Pests Using Color Based Image Segmentation" International Conference on Inventive Communication and Computational Technologies, 2018
29.	Barik R.C., Sahu S.S.,Bhoi S.P., Changder S. "A Novel Data Encryption Approach in the Grid-

	Structured Binary Image.", International Conference on Microelectronics, Computing & Communication Systems, Lecture Notes in Electrical Engineering, vol 453, pp:103-115, Springer, Singapore, 2018 (ISBN: 978-981-10-5564-5)
30.	Mishra A., Pandey C., Sahu S.S. "Detection of Atrial Arrhythmias Using Autoregressive Modeling" Innovations in Electronics and Communication Engineering. Lecture Notes in Networks and Systems, vol 7, pp: 193- 200, Nov. 2018, Springer, Singapore (ISBN: 978-981-10-3811-2)
31.	Navneet Nayan, Sanjeet Kumar and Sitanshu Sekhar Sahu "Accidental Event Detection Based on Optical Flow Analysis" Advances in Communication, Devices and Networking, Lecture Notes in Electrical Engineering vol. 462, pp: 607-614, May 2018, Springer, Singapore (ISBN: 978-981-10-7900-9)
32.	Sweta kumari, SitanshuSekharSahu and Bharat Gupta, "A Thyristor based Self Start up Integrated Circuit for Piezo-Electric Energy Harvester" International Conference on Intelligent Computing and Control, Coimbatore,23-24 June. 2017, pp: 94-98. (ISBN: 978-1-5386-0373-4)
33.	Swetakumari, SitanshuSekharSahu and Bharat Gupta, "A Low Voltage Pre Start Up Circuit for Thermal Energy Harvesting Applications" IEEE International Conference on Wireless Communications, Signal Processing and Networking (WiSPNET), Chennai, 22 -24 March, 2017, pp-1999-2002. (doi: 978-1-5090-4442-9/17)
34.	SprihaChandrayan, Aarushi Agarwal, Mohammad Arif, Sitanshu S Sahu,"Selection of dominant voice features for accurate detection of Parkinson's disease" IEEE International Conference on Biosignals, Images and Instrumentation (ICBSII), 2017, 7-8 March, Chennai, India (doi: 978-1-5090-4980-6/17)
35.	Ram Chandra Barik, SuvamoyChangder, Sitanshu Sekhar Sahu, "Image Texture-Based New Cryptography Scheme Using Advanced Encryption Standard" Computational Intelligence in Data Mining, Springer, Singapore, vol. 556, pp: 449-458, May 2017 (ISBN: 978-981-10-3873-0)
36.	R.K.Sinha and S.S.Sahu, "Quantum Key Distribution: Simulation of BB84 Protocol in C" International conference on Engineering Technology, Science and Management Innovation, 15th Jan, Chandigarh, India, 2017 (ISSN: 2345-117X)
37.	Ankita Mishra, Ashutosh Kumar Singh and SitanshuSekharSahu, "ECG Signal Denoising using Time-Frequency based Filtering Approach" IEEE International Conference on Communication and Signal Processing, Melmaruvathur, April 6-8, 2016.
38.	Aarushi Agarwal, SprihaChandrayan and Sitanshu S Sahu, "Prediction of Parkinson's Disease using Speech Signal with Extreme Learning Machine" IEEE International Conference on Electrical, Electronics, and Optimization Techniques (ICEEOT) , Chennai, March 3-5, 2016.
39.	Ashutosh Kumar Singh and Sitanshu Sekhar Sahu, "Prediction of host-pathogen protein-protein interaction network using Machine learning approach" National Conference on Emerging & Innovative Trends in Computer Science, Hyderabad, India, April 12-14, 2016. (ISBN:978-93-82570-78-3)
40.	Ashutosh Kumar Singh, Sitanshu S Sahu, Ankita Mishra, "Sub-Cellular Localization Prediction Using Machine Learning Approach" International Conference on Instrumentation, Electrical and Electronics Engineering, Raipur, India, Dec., 2015
41.	Sitanshu S. Sahu, Tyler Weirick, and Rakesh Kaundal, "Predicting genome-scale Arabidopsis-Pseudomonas syringaeinteractome using domain and interolog-based approaches", BMC Bioinformatics 15(S11): S13, 2014
42.	Tyler Weirick, Sitanshu S. Sahu, R. Mahalingam and Rakesh Kaundal, "LacSubPred: predicting subtypes of Laccases,an important lignin metabolism-related enzyme class, using in-silicoapproaches", BMCBioinformatics 15(S11): S15, 2014
43.	Sitanshu S. Sahu, AmitaKaundal, Clemencia Rojas, Seonghee Lee, Tyler Weirick, Kiran S. Mysore and Rakesh Kaundal,"Computational prediction and experimental validation of host-

	pathogen protein-protein interactome:a case study on Arabidopsis thaliana - Pseudomonas syringae model system”, Plant Biology, Portland, Oregon,USA,2014.
44.	Rakesh Kaundal, Sitanshu S. Sahu, Ruchi Verma and Tyler Weirick, “Identification and characterization of plastid-type proteins from sequence-attributed features using machine learning”, BMC Bioinformatics 14(S14): S7, 2013
45.	Sitanshu S. Sahu and Rakesh Kaundal. “Predicting Multi-target Protein Subcellular Localization combining homology and Machine Learning Approaches”, MCBIOS Conference, Columbia, MO, USA, 2013.
46.	Sitanshu S. Sahu, Ganapati Panda and Ramchandra Barik , “ Cancer Classification Using Microarray Gene Expression Data: Approached Using Wavelet Transform and F-score Method”, International Conference on Electronic Systems, NIT, Rourkela, Jan. 07-09, 2011.
47.	Sitanshu S. Sahu, Ganapati Panda, “An Efficient Signal Processing Approach in Eukaryotic Gene Prediction”, The Eighth Asia Pacific Bioinformatics Conference, Bangalore, India, Jan. 18-21, 2010
48.	Sitanshu S. Sahu, Ganapati Panda, Nithin V. George, “An Improved S-Transform for Time-Frequency Analysis”, IEEE International Advance Computing Conference (IACC), Thapar University, Patiala, India, pp: 315-319, 2009, DOI No: 10.1109/IADCC.2009.4809028
49.	Sitanshu S. Sahu, Ganapati Panda , “A New Approach for Identification of Hot Spots in Proteins Using S-Transform Filtering”, IEEE International Workshop on Genomic Signal Processing and Statistics (GENSIPS), Radisson University, Minnesota, USA, pp: 1-4, 2009, DOI No: 10.1109/GENSIPS.2009.5174374
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51.	Sitanshu S. Sahu, Ganapati panda and Satyasai Jagannath Nanda, “ Improved Protein Structural Class Prediction Using Genetic Algorithm and Artificial Immune System”, World Congress on Nature and Biologically Inspired Computing (NaBIC), Coimbatore, India, Dec.9-11, 2009, pp. 731-735, DOI No: 10.1109/NABIC.2009.5393488
52.	Sitanshu S. Sahu, Ganapati panda and Chinmaya Mahapatra, “Development of a new low complexity ANN based Drug discovery method for efficient design of SARS Co3CL Protease Inhibitors” ,International Congress of Environmental Research (ICER), BITS, Pilani, Vol.2, pp: 1044, Dec. 18-20, 2008
53.	Sitanshu S. Sahu and Ganapati panda , “Gene Prediction by Sliding DFT Algorithm”, International Conference Advance Bio-Engineering (ICAB), Satyabhama University, Dec. 05-07, 2007

Book Chapter

1.	Tuhin Mukherjee, Sitanshu Sekhar Sahu, Ashok Kumar Pattnaik, Kishanta Kumar Pradhan, Satyajit Mohanty, Navigating in silico drug discovery: A hands-on guide with AutoDock Vina and Discovery Studio, Harnessing Genomic Tools for Crop Improvement, Academic Press, Elsevier, 2026, ISBN: 978-0-443-29281-1 DOI: 10.1016/C2023-0-51864-6
2.	Pandit Vivek Kumar Pandey, Sitanshu Sekhar Sahu, Speech Signal Analysis Using Gammatone-Frequency Cepstral Coefficient For Parkinson's Disease Prediction, Book: Machine Learning in Biomedical and Health Informatics, pp:1-22, ISBN: 9781774919552, Apple Academic Press,

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3.	Biswajit Karan, Animesh Sharma, Sitanshu Sekhar Sahu, Sudhansu Kumar Mishra, Deep Neural Network for Parkinson Disease Prediction Using SPECT Image, Book: Deep Learning, Machine Learning and IoT in Biomedical and Health Informatics, Pages159-173, ISBN: 9780367548445,CRC Press, Taylor & Francis, Feb. 2022.
4.	Sudhansu Kumar Mishra, Prajna Parimita Dash, Sitanshu Sekhar Sahu, Ashutosh Rath, A Comparative Performance Assessment of a Set of Adaptive Median Filters for Eliminating Noise from Medical Images, Book: Deep Learning, Machine Learning and IoT in Biomedical and Health Informatics, Pages121-137, ISBN: 9780367548445, CRC Press, Taylor & Francis, Feb. 2022
5.	Rajeev Sharma, Sitanshu Sekhar Sahu, Abhay Upadhyay, Rishi Raj Sharma, and Ajit Kumar Sahoo, "Sleep Stage Classification Using DWT and Dispersion Entropy Applied on EEG Signals" Computer-Aided Design for Biomedical Applications, CRC Press, Taylor & Francis Group, pp:35-56, April 2021 ISBN: 9781003121152
6.	Sweta Kumari, Sitanshu Sekhar Sahu, Bharat Gupta, Sudhansu Kumar Mishra, "Energy harvesting via human body activities" Smart Biosensors in Medical Care, Academic Press, Elsevier, pp: 87-106, 2020, ISBN: 978-0-12-820781-9
7.	Biswajit Karan, Kartik Mahto and Sitanshu Sekhar Sahu , "Intelligent Speech Processing in the Time-Frequency Domain", Intelligent Speech Signal Processing, Elsevier, PP: 153-169. 2019, ISBN: 978-0-12-818130-0

Funded Project

Completed:

- 1- Title: Development of end-to-end speech-based AI system for Parkinson's Disease Prediction; IDEAS, TIH, ISI Kolkata, 2022-2025, amount: 5,14,000 (PI)
- 2- Title: Computational identification and validation of genome wide protein-protein interaction network in rice and rice blast; SERB, DST, Govt. of India, 2017-2020, amount: 21,62,600. (PI)
- 3- Title: Electroencephalogram based Sleep Stages Detection using Advance Wavelet Techniques; NPIU, MHRD, Govt. of India, 2019-2021, amount: 14,42,000 (PI)
- 4- Title: Development of Data Science Methods for Cyber-Physical System, Funded by NMICPS, TIH, ISI Kolkata DST, 2022-2025 Amount: 1,00,00,000 (PI)

Patent

- 1- A photo biomodulation device for attenuating obesity, and method thereof (Granted) (Indian patent No. 564886)
- 2- An intelligent irrigation system and method thereof (Granted) (Indian Patent No. 599883)
- 3- A piezoelectric based energy harvesting system and method thereof (Published) (Application No. 202331057064)
- 4- Agricultural Health Monitoring Device (Granted), UK Design Patent (No. 6300570)

- 5- A System for Real Time Detection of Parkinson's Disease, Method and Utility (Application No: 202531093457)
- 6- An Autonomous Vehicle Navigation System and Method Thereof (Published) (Application No: 202331066870, October 5, 2023)
- 7- A DEVICE FOR HEART CONDITION(S) MANAGEMENT AND METHOD THEREOF, (Application no. 202431024417 Dt:27/03/2024)

Ph.D Guidance

- 1- Pandit Vivek Kumar Pandey, "Multilingual Speech Signal Analysis for Parkinson's Disease Detection" 2026 (Awarded)
- 2- Tuhin Mukherjee, "Investigation of RRM based Low Level laser Therapy for Obesity and Obesity-induced cardiac Ischemic Complications" 2026 (Awarded)
- 3- Satyajit Mahapatra: "Machine Learning and Signal Processing Techniques for Protein-Protein Interaction Prediction" 2022 (Awarded)
- 4- Biswajit Karan: "Analysis of Speech Signal for Parkinson's Disease Prediction" 2021 (Awarded)
- 5- Sweta Kumari: "Design and Implementation of Controller for Energy Harvesting Applications" 2020 (Awarded)
- 6- Rupesh Kumar Sinha: "Chaos based Symmetric Cryptography for Image Security" 2019 (Awarded)

Workshop Organized

1. "Artificial Intelligence and Machine learning in Healthcare", 27th -31st July 2020 at BIT Mesra sponsored by TEQIP-III
2. "Artificial Intelligence" 25th -29th January 2021 at BIT Mesra sponsored by ATAL-AICTE.
3. "Machine Learning in Speech and Audio Processing", 21st-25th Feb. 2022 at BIT Mesra
4. Conducted workshop on "Embedded AI Design" 20th-24th February 2023
5. Conducted workshop on "Internet of Things (IoT)" 25th-26th April 2023
6. "Recent Trends in in Speech and Audio Processing", 16th-20th March. 2026 at BIT Mesra



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